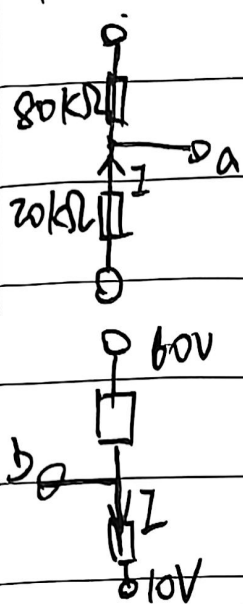


[2-12]



$$U_{aE} = \varphi_a - (-20V) = \frac{40V}{100k\Omega} \cdot 80k\Omega$$

$$\Rightarrow \varphi_a = 12V$$

$$U_{bE} = \varphi_b - 10V = \frac{50V}{10k\Omega} \cdot 8k\Omega$$

$$\therefore \varphi_b = 50V$$

$$U_{gk} = U_s + IR_B$$

$$\therefore U_E = -\mu U_{gk} - IR_B = -\mu U_s - (\mu + 1)IR_B$$

$$U_L = -\mu U_{gk} - IR_B = -\mu U_s - (\mu + 1)IR_B = IR$$

$$\text{解得 } I = \frac{-\mu U_s}{(\mu + 1)R_B + R_L}$$

$$\therefore |U_L| = \frac{\mu U_s R_L}{(\mu + 1)R_B + R_L} \quad \text{上正下负}$$

[2-18] (1) $R_c = \frac{U_c}{I_c} = 2 \times 10^3 \Omega$

emmm. U_c 指的是 C 点电势?

(2) $U_{CE} = -U_c + V_{CC} - U_E = 1V$

表述不清吧

$$U_{BE} = I_2 R_2 - U_E = 0.65V$$

$$R_c = \frac{V_{CC} - U_c}{I_c} = 1000 \Omega$$

(3) $I_1 = I_B + I_2 = 200 \mu A$

$$\begin{cases} I_E + I_2 = I' \\ I' = I_1 + I_c \end{cases}$$

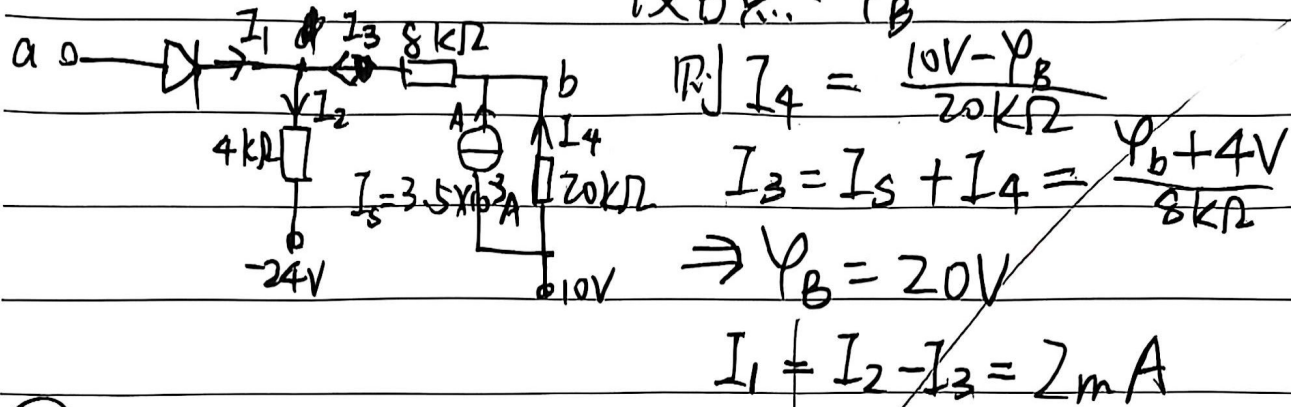
0.2mA (当成 2mA, 太急了)

$$\Rightarrow I_E = I_1 + I_c - I_2 = 3.85mA$$

12-20

① $V_a = -4V$ 时, 短路: $I_2 = \frac{20V}{4k\Omega} = 5 \times 10^{-3} A$

设 B 点: V_B



② ~~即: 使 $V_a = 0$~~

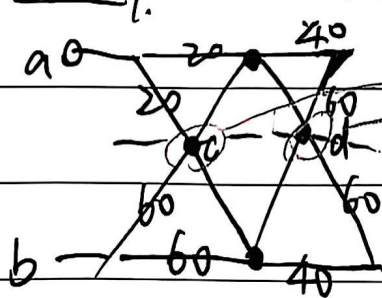
即: 令有 $I_1 = 0, I_2 = I_3$

$$I_2 = \frac{V_a + 24V}{4k\Omega}, \quad I_3 = 3.5mA + \frac{10V - V_B}{20k\Omega}$$

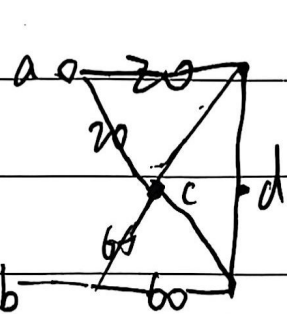
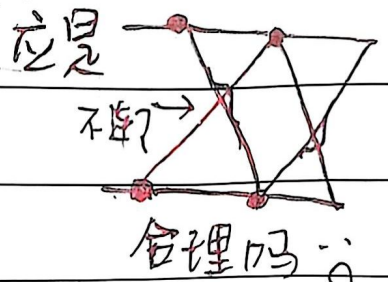
I_3 与 ① 中不变, $I_3 = 3mA$

$$\therefore V_a = -12V$$

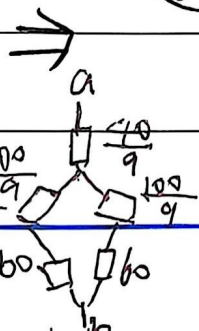
4-2 ① $R = 20\Omega$ 对称, $V_c = V_d$ 短路



靠, 这两点不相通



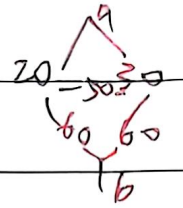
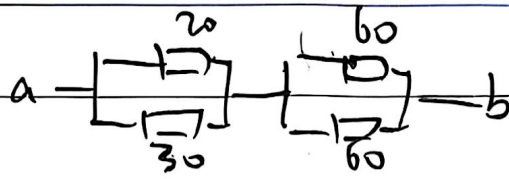
$$V_c = V_d$$



$$R_{ab} = 40\Omega$$

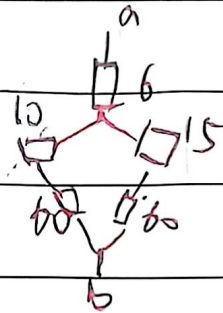
$$R_{ab} = \frac{40}{9} + \frac{1}{\frac{1}{20} + \frac{1}{60}} = 40\Omega$$

② $R = 30\Omega$



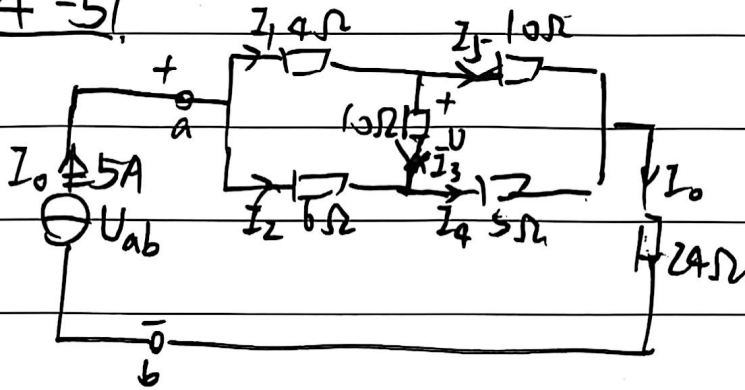
$R_{ab} = 42\Omega$

$\Delta \rightarrow Y$



$R_{ab} = 6 + \frac{1}{\frac{1}{10} + \frac{1}{15}} = 42\Omega$

4-5



$I_0 = I_1 + I_2$

$I_1 + I_3 = I_5$

$I_2 = I_3 + I_4$

$I_2 \cdot 6\Omega + I_4 \cdot 5\Omega + I_0 \cdot 24\Omega = U_{ab}$

$I_1 \cdot 4\Omega + I_5 \cdot 10\Omega + I_0 \cdot 24\Omega = U_{ab}$

$I_1 \cdot 4\Omega - U + I_4 \cdot 5\Omega + I_0 \cdot 24\Omega = U_{ab}$

$U = I_3 \cdot 10\Omega$

联立: 有

$I_3 = 0.5A$

$I_5 = 2A$

$I_4 = 3A$

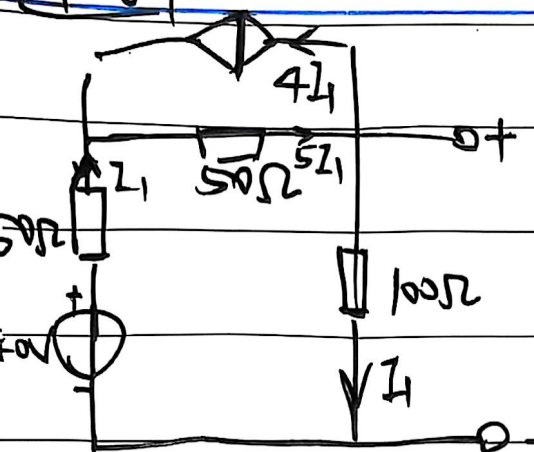
$I_2 = 2.5A$

$I_1 = 2.5A$

$U = 5V$

$U_{ab} = 150V$

14-8



$$U = I_1 \cdot 100\Omega$$

$$U = -5I_1 \cdot 50\Omega + I_1 \cdot 50\Omega + 40V$$

得 $I_1 = \cancel{0.1} 0.1 A$

$U = \cancel{5} 10 V$